

Q.1. Answer the following short questions:

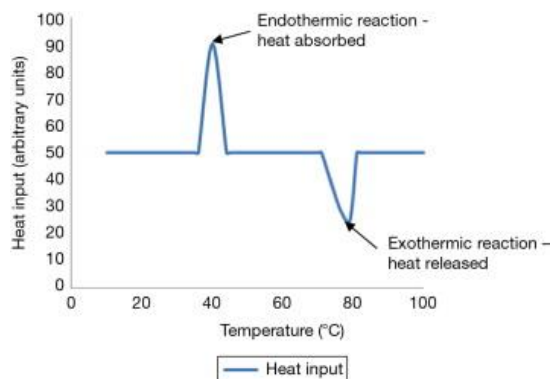
(15X2=30)

1. Differentiate between DTA and DSC.

DTA	DSC
Measures the temperature difference between the sample and a reference.	Measures the heat flow into or out of a sample compared to a reference.
Less sensitive to small thermal events.	More sensitive, can detect small thermal events.
Typically, at the same temperature.	Sample and reference may not be at the same temperature.

2. What is DTA curve? Give its characteristics.

A **DTA** curve is a graphical representation obtained from a Differential Thermal Analysis (DTA) experiment. It plots the temperature difference between a sample and a reference material (ΔT) as a function of the temperature of the sample or time.



Characteristics:

- **Baseline:** A flat line indicating no thermal difference between the sample and the reference.
- **Endothermic Peaks:** Downward peaks that represent heat absorption by the sample (e.g., melting or vaporization).

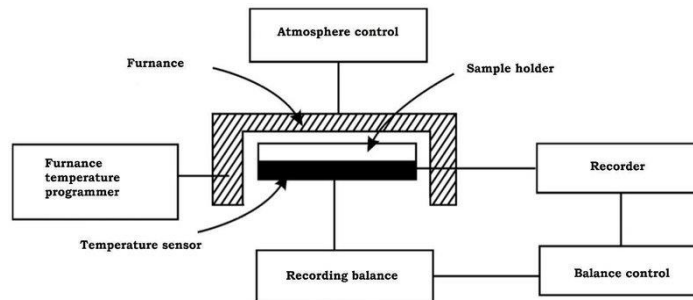
- **Exothermic Peaks:** Upward peaks that represent heat release by the sample (e.g., crystallization or decomposition).

3. Describe the thermal analysis?

Thermal analysis is a method used to study how materials change in response to temperature variations. By monitoring properties such as heat flow, mass loss, or dimensional changes during heating or cooling, it provides insights into a material's thermal stability, phase transitions, and chemical reactions.

- Differential Scanning Calorimetry (**DSC**)
- Differential Thermal Analysis (**DTA**)
- Thermogravimetric Analysis (**TGA**)

4. Write down the different components of thermobalance.



- **Balance:** Measures the mass of the sample with high precision.
- **Furnace or Heating/Cooling Unit:** Provides controlled heating or cooling to the sample.
- **Sample Pan:** Holds the sample during the thermal analysis.
- **Reference Pan:** Holds a reference material, used to account for background changes.
- **Thermocouple or Temperature Sensor:** Monitors the temperature of the sample and furnace.
- **Control Unit:** Manages the temperature programming and data acquisition.
- **Data Acquisition System:** Collects and records the mass changes and temperature data.
- **Software:** Analyzes the data and generates thermogravimetric curves for interpretation.

5. What is common source of errors in thermogravimetric analysis?

- **Sample Preparation:** Inconsistent sample size or improper sample placement can affect results.
- **Thermal Drift:** Gradual changes in temperature due to equipment instability can impact accuracy.
- **Environmental Contamination:** Exposure to moisture or air can cause sample weight changes unrelated to the actual thermal process.
- **Instrumental Noise:** Electrical or mechanical noise can interfere with precise weight measurements.
- **Incomplete Sample Combustion:** Residual materials or incomplete reactions can lead to erroneous mass loss readings.

6. What is an enthalpogram?

An **enthalpogram** is a graphical representation of the heat flow versus temperature or time, obtained from a Differential Scanning Calorimetry (DSC) experiment. It provides a visual depiction of the enthalpy changes occurring in a sample as it undergoes thermal events such as melting, crystallization, or chemical reactions. The enthalpogram typically shows peaks and valleys that correspond to endothermic (heat absorption) and exothermic (heat release) processes. By analyzing the enthalpogram, one can determine the thermal properties of a material, including phase transitions and reaction enthalpies.

7. Discuss the applications of Differential Scanning Calorimetry.

- **Measuring Heat Capacities:** DSC helps determine the specific heat capacities of materials, which provides insight into their thermal behavior and stability.
- **Analyzing Phase Transitions:** It detects various phase transitions, such as glass transitions and crystallization, helping understand the thermal properties of polymers and other materials.
- **Studying Thermal Stability:** DSC assesses the decomposition temperatures of materials, providing information about their stability under thermal conditions.

- **Pharmaceutical Analysis:** DSC is used to analyze drug formulations for purity, stability, and potential interactions, ensuring the quality and efficacy of pharmaceutical products.
- **Quality Control:** In manufacturing, DSC ensures consistency and quality by monitoring the thermal properties of materials and products.

8. What are the common instrumental factors affecting TG curves?

- **Sample Mass:** Inconsistent or excessively large sample masses can lead to uneven heating and inaccurate mass loss data.
- **Heating Rate:** The speed at which the sample is heated can affect the resolution and shape of the TG curve, with faster rates potentially missing subtle transitions.
- **Atmosphere:** The type of gas (e.g., nitrogen, oxygen) and its flow rate can influence the thermal decomposition of the sample and the TG curve shape.
- **Temperature Calibration:** Incorrect calibration of the temperature sensor can lead to errors in determining the temperatures at which thermal events occur.
- **Sample Holder Material:** The material and design of the sample holder can impact heat transfer and thermal conductivity, affecting the accuracy of the TG measurements.

9. What is a thermocouple?

A **thermocouple** is a temperature sensor made from two different metals joined at one end. When the junction of these metals is exposed to a temperature difference, it generates a small voltage that is proportional to the temperature difference. This voltage can be measured and converted into a temperature reading. Thermocouples are widely used for their simplicity, durability, and ability to measure a wide range of temperatures. They are commonly used in various applications, including industrial processes and scientific experiments.

10. What are the applications of evolved gas detection (EGD)?

- **Thermal Analysis:** Identifying and quantifying gases released during the heating of samples, such as decomposition products, which helps in understanding material stability and composition.
- **Material Characterization:** Analyzing gases evolved during chemical reactions to determine the presence of specific compounds, aiding in the detailed characterization of materials.
- **Reaction Kinetics:** Studying the rate and mechanism of chemical reactions by monitoring the amount and type of gas released, providing insights into reaction pathways and conditions.
- **Quality Control:** Ensuring the quality and consistency of materials by detecting and analyzing gases released during manufacturing processes, which helps in maintaining product standards.

11. Discuss the limitations of DSC?

- **Sample Size Constraints:** DSC typically requires small sample sizes, which may not be suitable for all types of analyses or large quantities of material.
- **Limited to Thermal Processes:** DSC primarily detects thermal events such as melting, crystallization, and reactions. It may not be effective for analyzing non-thermal properties or processes.
- **Sample Preparation:** Some samples require specific preparation methods to ensure accurate results, which can be time-consuming and complex.
- **Heat Flow Accuracy:** The accuracy of heat flow measurements can be affected by baseline drift, instrumental noise, or inadequate calibration, leading to potential errors in data interpretation.

12. DTA and DSC, which method is preferred for quantitative purposes and why?

DSC (Differential Scanning Calorimetry) is preferred for quantitative purposes over **DTA (Differential Thermal Analysis)**. This is because DSC measures the exact heat flow difference

between the sample and a reference material, providing precise quantitative data on thermal events such as heat capacities, enthalpy changes, and phase transitions. DSC offers detailed and accurate measurements of thermal properties by directly quantifying the heat absorbed or released, which allows for more reliable and specific analysis of material behaviors compared to DTA, which only indicates temperature differences and does not provide detailed heat flow information.

13. Describe the Heat Flux DSC.

- **Heaters** are employed to heat the substance inside the DSC, with the amount of heat controlled by a computer.
- **Two compartments** are present: one for the sample and one for the reference material.
- **Thermocouples** made of Chromel (a metal alloy) are used to measure the temperature of both the sample and the reference.
- The **heat difference** is measured using a Chromel-Alumel junction.
- An **amplifier** is used to amplify the signal, and the results are recorded in the form of a graph.

14. What are the advantages of DSC.?

- **High Sensitivity:** Can detect small heat changes, making it suitable for analyzing subtle thermal events.
- **Quantitative Analysis:** Provides precise measurements of heat flow, allowing for accurate calculation of heat capacities and enthalpy changes.
- **Versatility:** Can be used for a wide range of materials, including polymers, pharmaceuticals, and food products.
- **Detailed Thermal Information:** Helps in identifying and characterizing phase transitions, such as melting, crystallization, and glass transitions.

15. Discuss the limitations of TGA?

- TGA generally requires small sample sizes, which may not be suitable for all types of analyses or larger quantities of material.
- The temperature range of TGA instruments may limit the analysis of materials that decompose or transition outside this range.
- TGA can only detect mass changes, so if a sample undergoes complex reactions or decomposes into gases that are not captured, it may not provide complete information.

Answer the following question:

(3X10=30)

Q. No. 2:

- (a) Discuss the instrumentation of Differential Thermal Analysis (DTA).
- (b) Write down the analytical applications of Thermometric Titrations.

Q. No. 3:

- (a) Describe the basic principle of Differential Scanning Calorimetry (DSC).
- (b) Briefly, discuss instrumentation of Thermometric Titrations.

Q. No. 4:

- (a) Write down the basic principle of Evolved Gas Detection (EGD).
- (b) Discuss applications of Thermogravimetric Analysis (TGA).

From BS Regular

Q.1. Answer the following short questions:

(11X2=22)

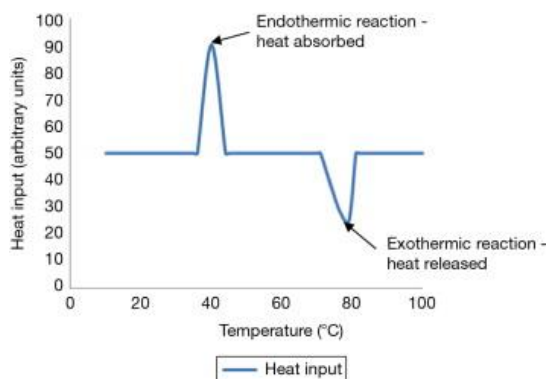
1. Write down the principle of TGA and DTA.

Principle of TGA: Measures mass changes of a sample as it is heated or cooled to determine thermal stability and composition. The mass loss or gain is recorded and plotted against temperature.

Principle of DTA: Measures the temperature difference between a sample and a reference as both are heated or cooled. This difference, plotted against temperature, reveals endothermic or exothermic thermal events.

2. What is DTA curve give its characteristics.

A **DTA** curve is a graphical representation obtained from a Differential Thermal Analysis (DTA) experiment. It plots the temperature difference between a sample and a reference material (ΔT) as a function of the temperature of the sample or time.



Characteristics:

Baseline: A flat line indicating no thermal difference between the sample and the reference.

Endothermic Peaks: Downward peaks that represent heat absorption by the sample (e.g., melting or vaporization).

Exothermic Peaks: Upward peaks that represent heat release by the sample (e.g., crystallization or decomposition).

3. How TGA can be used to evaluate the thermal stability of a material?

TGA evaluates thermal stability by measuring weight loss of a material as it is heated. As the temperature increases, the material's weight decreases due to decomposition or volatilization. The TGA curve shows the temperature at which weight loss begins and the rate of weight loss, indicating the material's thermal stability and decomposition points. This information helps assess how well the material can withstand high temperatures.

4. What is the basic theory of DTA?

DTA measures the temperature difference between a sample and a reference as both are subjected to the same heating or cooling conditions. As the sample undergoes thermal changes like melting or crystallization, it absorbs or releases heat differently than the reference. This difference in heat flow is recorded as a temperature difference and plotted against temperature. Peaks in the DTA curve indicate endothermic processes (heat absorption) or exothermic processes (heat release), providing insights into phase transitions and thermal properties of the material.

5. What is principle of TGA?

TGA measures how a sample's mass changes as it is heated or cooled. By continuously recording the sample's weight, TGA detects mass loss or gain due to thermal events like decomposition, oxidation, or volatilization. The data is plotted as a mass versus temperature curve, which shows at what temperatures these changes occur. This information helps evaluate the sample's thermal stability, composition, and reaction behavior.

6. Differentiate between heat flux DSC and power compensated DSC.

Heat Flux DSC	Power Compensated DSC
Measures the heat flow difference between a sample and a reference using heat flux sensors.	Measures the power needed to maintain the same temperature for both the sample and reference.
Heat flow is detected as the sample and reference are heated or cooled together.	Power is adjusted to keep the sample and reference at the same temperature, and the difference in power is measured.

7. Give two differences between DSC and DTA.

DTA	DSC
Measures the temperature difference between the sample and a reference.	Measures the heat flow into or out of a sample compared to a reference.
Less sensitive to small thermal events.	More sensitive, can detect small thermal events.

8. What is thermal analysis?

Thermal analysis is a method used to study how materials change in response to temperature variations. By monitoring properties such as heat flow, mass loss, or dimensional changes during heating or cooling, it provides insights into a material's thermal stability, phase transitions, and chemical reactions.

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10. Define thermometric titration.

Thermometric titration is a method where temperature changes are measured during a titration to determine the endpoint of the reaction. As the titrant is added to the analyte, any heat released or absorbed causes a temperature change, which is monitored using a temperature sensor. The endpoint is identified from the temperature change data plotted against the volume of titrant added.

Answer the following question:

(5X5=25)

- a) Write down the application of differential scanning calorimetry (DSC).
- b) Discuss the factors that affecting the differential thermal analysis.
- c) Describe Instrumentations of TGA with the help of a Diagram.
- d) Describe application of TGA curve.
- e) Write a note on instrumentation used for thermometric titration.